

A study of the quantum harmonic oscillator in the path integral formalism via MCMC

Tommaso Serra

11 February 2026

1 Introduction

This report presents a simple study of the quantum harmonic oscillator by means of Markov Chain Monte Carlo (MCMC) simulations, using the path integral formulation. We estimate the thermal averages of a few relevant physical observables and the first energy gaps, making use of suitable interpolating operators and of their connected correlators. Finally, we reconstruct the squared modulus of the ground-state wave function and compare it with the analytic solution.

1.1 The harmonic oscillator

The Hamiltonian of the harmonic oscillator is

$$H = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2}m\omega^2 x^2 \quad (1)$$

whose eigenstates in the position representation are

$$\psi_n(x) = \frac{1}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} e^{-\frac{m\omega x^2}{2\hbar}} H_n \left(\sqrt{\frac{m\omega}{\hbar}} x \right) \quad (2)$$

where the H_n are the Hermite polynomials. The corresponding energy eigenvalues are

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right) \quad (3)$$

1.2 Quantum statistics and the Euclidean path integral

The partition function of a quantum system is computed as

$$Z = \text{Tr}(e^{-\beta H}) \quad (4)$$

To make the connection with the path integral for real-time propagation explicit, it can be written as

$$Z = \int dx \langle x | e^{-\beta H} | x \rangle \quad (5)$$

We can apply the same path integral procedure by performing the substitution $t \rightarrow -i\tau = -i\beta$, moving to imaginary time through the Wick rotation (the only difference lies in the need to impose periodic boundary conditions). In this formulation the partition function, in the continuum limit, becomes (with $\hbar = 1$)

$$Z \propto \int_{x(0)=x(\beta)} \mathcal{D}x(\tau) e^{-S_E[x(\tau)]} \quad (6)$$

where S_E is the Euclidean action, given by

$$S_E[x(\tau)] = \int_0^\beta d\tau' \left[\frac{1}{2}m \left(\frac{dx}{d\tau'} \right)^2 + V(x(\tau')) \right] \quad (7)$$

At this point we introduce the thermodynamic average of an operator O :

$$\langle O \rangle = \sum_n P_n O_n \quad (8)$$

where

$$P_n = \frac{e^{-\beta E_n}}{Z} \quad O_n = \langle n | O | n \rangle$$

so that

$$\langle O \rangle = \frac{\text{Tr}(O e^{-\beta H})}{Z} \quad (9)$$

Applying the path integral procedure to the numerator in the same way, we obtain

$$\langle O \rangle = \frac{\int \mathcal{D}x(\tau) e^{-S_E[x(\tau)]} O[x(\tau)]}{\int \mathcal{D}x(\tau) e^{-S_E[x(\tau)]}} \quad (10)$$

Thermal averages are therefore obtained as averages over all periodic paths, weighted by the exponential of the Euclidean action. We can estimate these values by designing a Markov chain that samples the space of paths according to the above distribution. In the numerical simulation the paths are discretised into N_t time slices, and the continuum limit can be analysed by decreasing the time step $\frac{\beta}{N_t}$ at fixed β .

2 Markov chain

The discrete Euclidean action for a path x_i with $i = 0, 1, \dots, N-1$ is

$$S_E^{(L)}[x_i] = a \sum_{i=0}^{N_t-1} \left[\frac{m\omega^2 x_i^2}{2} + \frac{m}{2} \left(\frac{x_{i+1} - x_i}{a} \right)^2 \right] \quad (11)$$

It is convenient to move to dimensionless units by setting

$$\eta = a\omega \quad y_i = \frac{x_i}{l} \quad l = \sqrt{\frac{1}{m\omega}} \quad (12)$$

where $\frac{1}{\omega}$ is the characteristic time scale of the system. The action then becomes

$$S_E^{(L)} = \sum_{i=0}^{N_t-1} \left[y_i^2 \left(\frac{\eta}{2} + \frac{1}{\eta} \right) - \frac{1}{\eta} y_i y_{i+1} \right] \quad (13)$$

The probability distribution to be sampled is therefore

$$P(y_0, \dots, y_{N_t-1}) = \frac{\exp \left\{ - \sum_{i=0}^{N_t-1} \left[y_i^2 \left(\frac{\eta}{2} + \frac{1}{\eta} \right) - \frac{1}{\eta} y_i y_{i+1} \right] \right\}}{\int \prod_{j=0}^{N_t-1} dy_j \exp \left\{ - \sum_{i=0}^{N_t-1} \left[y_i^2 \left(\frac{\eta}{2} + \frac{1}{\eta} \right) - \frac{1}{\eta} y_i y_{i+1} \right] \right\}} \quad (14)$$

We implemented a chain made of 1 heat-bath sweep and 5 over-relaxation (microcanonical) sweeps which, combined together, guarantee aperiodicity and irreducibility.

2.1 Heat-bath update

The local heat-bath algorithm determines the new value y_i^t of the path independently of the value y_i at the previous step, by sampling directly from the conditional distribution $P(y_i | y_{j \neq i})$. In our case this algorithm is particularly efficient, since that distribution is simply a Gaussian centred at

$$\mu_i = \frac{\gamma}{2\alpha} \quad \gamma = \frac{y_{i-1} + y_{i+1}}{\eta} \quad \alpha = \frac{\eta}{2} + \frac{1}{\eta} \quad (15)$$

which depends only on the values of the adjacent points y_{i-1} and y_{i+1} . The variance is

$$\sigma^2 = \frac{1}{2\alpha} \quad (16)$$

The acceptance for the new state is equal to 1. The Box-Muller algorithm is used to sample this Gaussian.

2.2 Microcanonical update

The microcanonical (over-relaxation) update is a deterministic update that preserves the Euclidean action, so that the acceptance probability is always 1. The new value y_i^t is computed as

$$y_i^t = 2\mu - y_i \quad (17)$$

where μ is the conditional mean defined above (equation 15). That is, we are reflecting the point y_i about the mean of the Gaussian. This kind of update is particularly effective at reducing autocorrelations between samples and is computationally cheap.

3 Simulation and data collection

3.1 Simulation parameters

In order to study the behaviour of the energy as a function of temperature and to verify the agreement with theory, we fixed $\eta = 0.1$ and varied $\beta\hbar\omega$ from 1 to 20. We then fixed $\beta\hbar\omega = 10$, using values of N_t from 4 to 200, in order to study the continuum limit of the observables and the first three energy gaps, estimated through the connected correlators of y , y^2 , y^3 and A (using only the largest values of N_t).

We fixed the number of Markov chain steps to 10^7 , sampling the observables every 10 steps. The particular combination of heat bath, over-relaxation and sampling frequency resulted in very short autocorrelation times (of order unity, up to a maximum of a few tens) and a very small τ_{exp} . We chose a hot start, initialising the path with random values drawn from a uniform pdf on $[-1, 1]$ and discarding, for safety, 5×10^5 thermalisation steps.

3.2 Measured observables

We measured $\langle H \rangle$, $\langle y \rangle$, $\langle y^2 \rangle$, $\langle y^3 \rangle$ and $\langle A \rangle$, where A is the interpolating operator of the third excited state of the oscillator, defined by

$$A = y^3 - \frac{3}{2}y \quad (18)$$

We also measured the connected correlators

$$C_O(n) = \langle O(n)O(0) \rangle - |\langle O \rangle|^2 \quad (19)$$

for $n = 0, 1, \dots, \frac{N_t}{2}$, for the operators $\langle y \rangle$, $\langle y^2 \rangle$, $\langle y^3 \rangle$ and $\langle A \rangle$.

3.3 Estimators used

The estimates of $\langle y \rangle$, $\langle y^2 \rangle$, $\langle y^3 \rangle$ and $\langle A \rangle$ are obtained as simple averages of the corresponding observables (which depend only on the position) over the sampled paths,

$$\bar{y} = \frac{1}{N_t} \sum_{i=1}^{N_t} y_i, \quad \bar{y^2} = \frac{1}{N_t} \sum_{i=1}^{N_t} y_i^2, \quad \bar{y^3} = \frac{1}{N_t} \sum_{i=1}^{N_t} y_i^3, \quad \bar{A} = \frac{1}{N_t} \sum_{i=1}^{N_t} \left(y_i^3 - \frac{3}{2}y_i \right) \quad (20)$$

The expectation values of y , y^3 and A all vanish by symmetry. The energy is computed as

$$\bar{H} = \left\langle \frac{1}{2N_t} \sum_{i=0}^{N_t-1} y_i^2 + \frac{1}{2\eta} - \frac{1}{2N_t} \sum_{i=0}^{N_t-1} \frac{(y_{i+1} - y_i)^2}{\eta^2} \right\rangle \quad (21)$$

to be compared with its theoretical value which, in dimensionless units, coincides with the value of $\langle y^2 \rangle$ and reads

$$\langle H \rangle = \langle y^2 \rangle = \frac{1}{2} \frac{1 + e^{-\beta\hbar\omega}}{1 - e^{-\beta\hbar\omega}} \quad (22)$$

The estimates of the correlators are obtained from

$$\overline{O(n)O(0)} = \frac{1}{N_t} \sum_{i=0}^{N_t-1} O(x_i)O(x_{i+n}) \quad (23)$$

4 Analysis and results

4.1 Uncertainty estimation

The uncertainties on the primary observables were estimated with the blocking procedure. The uncertainties on the connected correlators and on the energy gaps were estimated using a blocked jackknife, looking for the plateau in order to obtain a reliable value while handling the autocorrelation between the data. As an illustration, figures 1 and 2 show the behaviour of the uncertainties as a function of the block size, where the plateau is clearly visible both for the blocking on the primary observables and for the jackknife on the connected correlators.

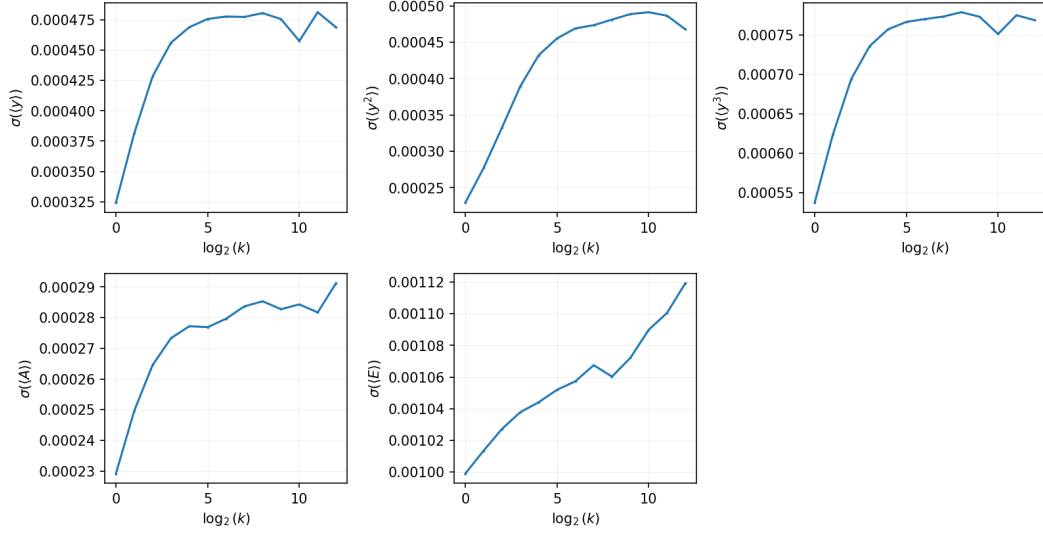


Figure 1: Behaviour of the uncertainties on $\langle y \rangle$, $\langle y^2 \rangle$, $\langle y^3 \rangle$, $\langle A \rangle$ and $\langle E \rangle$ as a function of the block size, for $\beta\hbar\omega = 10$ and $N_t = 200$.

The uncertainty on the energy appears to saturate less clearly. In fact the spread of the values is relatively small but, because of the zoom on the vertical axis, it appears amplified.

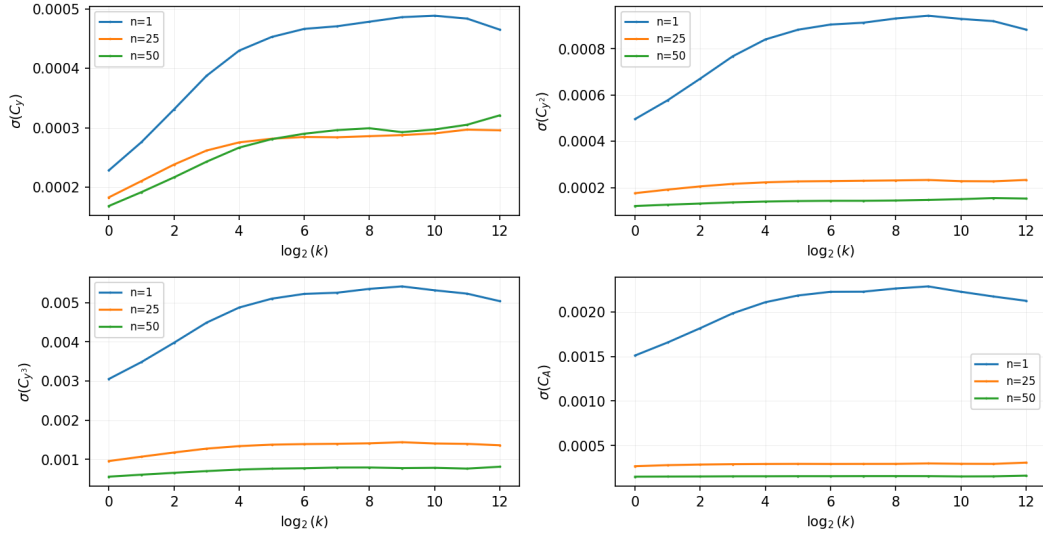


Figure 2: Behaviour of the uncertainties on the connected correlators for $n = 1, 25, 50$ (as an example) as a function of the block size, for $\beta\hbar\omega = 10$ and $N_t = 200$.

4.2 Energy as a function of temperature

Figure 3 shows the behaviour of the energy as a function of temperature.

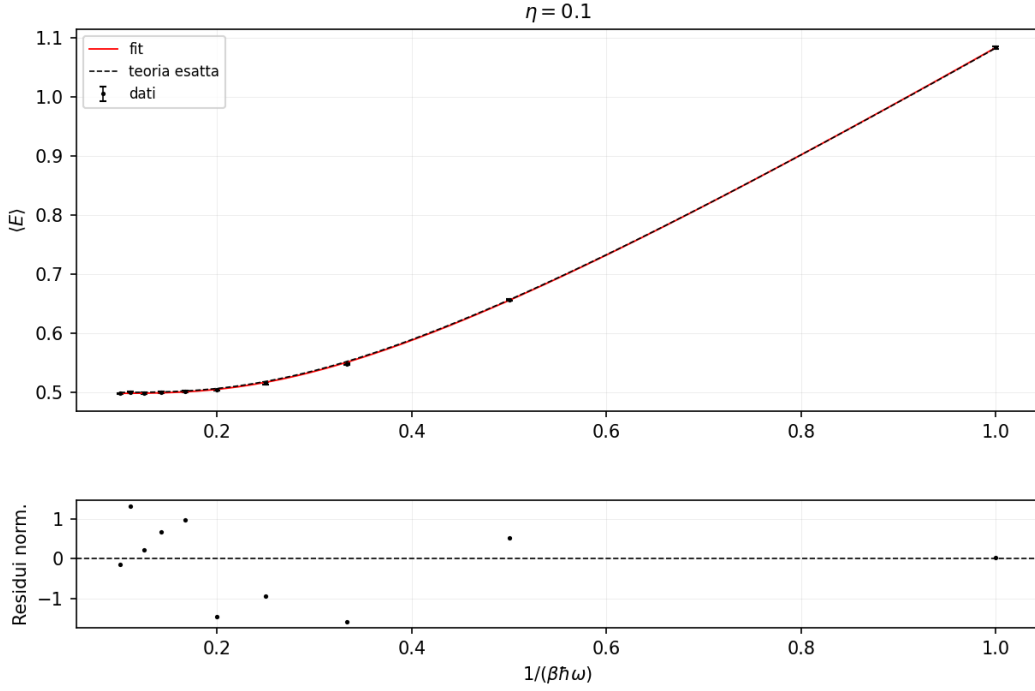


Figure 3: Behaviour of the energy as a function of temperature for $\eta = 0.1$.

We performed a two-parameter fit with the model

$$a + \frac{b}{e^{\beta\hbar\omega} - 1} \quad (24)$$

which corresponds to the theoretical expression of the energy (equation 22), with a expected to give the ground-state energy $E_0 = 0.5$ and b expected to equal $\hbar\omega = 1$.

Table 1: Fit results.

a	0.4985(4)
b	1.004(4)
χ^2_ν	1.144
ndf	8
$corr(a, b)$	-0.272

It should be noted that this procedure is not rigorous, since the energy data used are not the appropriate extrapolations to the continuum limit. Having fixed the step η exposes us to finite-step deviations of order $O(\eta^2)$, and probably to values of N_t that are too small for the smallest values of $\beta\hbar\omega$. For a more reliable result one could adopt the approach described in the following section, using several values of N_t for each $\beta\hbar\omega$ and performing a linear fit in η^2 :

$$E(\eta^2) = a + b\eta^2 \quad (25)$$

In this way one extrapolates the continuum limit of the energy for each value of $\beta\hbar\omega$, and the fit as a function of temperature can be performed using those values, corresponding to the intercept of the linear fit. Because of limited computational resources and insufficient statistics, we proceeded as described above anyway, assuming that $\eta = 0.1$ is small enough not to introduce significant deviations from the continuum-limit model. The fit gives a value of a that differs from the theoretical value of 0.5 by almost 4 sigma, while b is well compatible (table 1).

4.3 Finite-step effects and continuum extrapolation

Here we carry out the analysis mentioned above for a single value of $\beta\hbar\omega$. Figure 4 also shows the behaviour of $\langle y \rangle$ and $\langle y^2 \rangle$ for $\beta\hbar\omega = 10$.

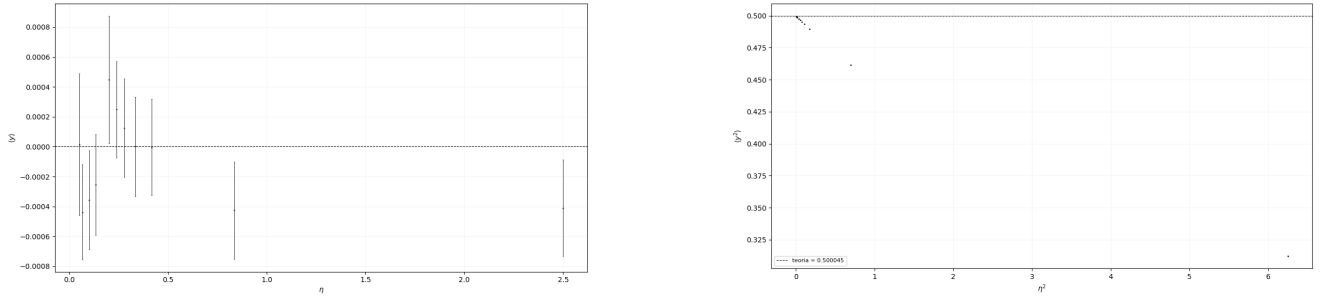


Figure 4: Values of $\langle y \rangle$ and $\langle y^2 \rangle$ for $\beta\hbar\omega = 10$ as a function of N_t : y is compatible with zero, while y^2 appears compatible with the theoretical value of ≈ 0.5 in the continuum limit.

We now show the plot of the energy (figure 5). We performed a linear fit in η^2 with the model of equation (25), considering the values with $\eta < 0.5$, in order to test the continuum limit and to capture the $O(\eta^2)$ deviations due to finite-step effects (higher-order deviations are also visible for the larger values of η , which are not included in the fit).

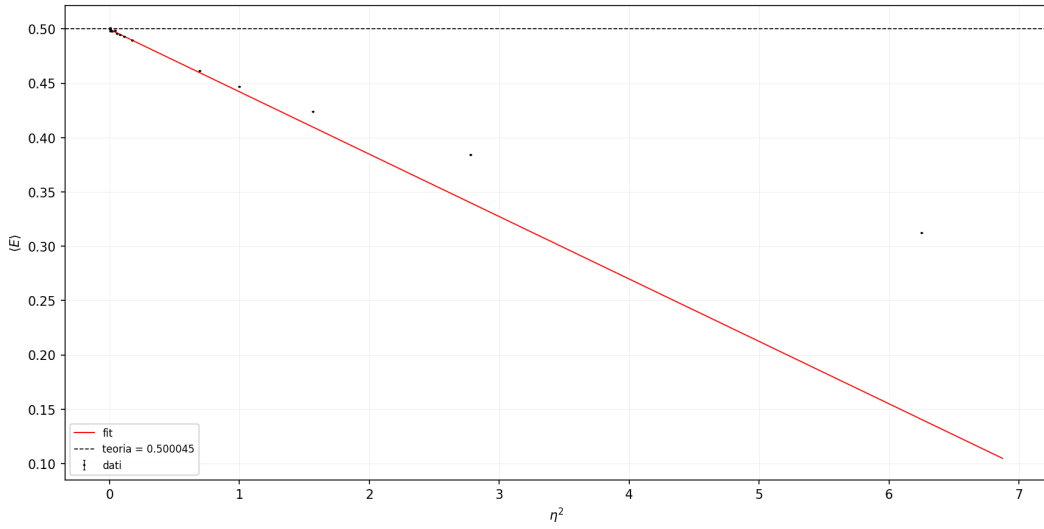


Figure 5: Behaviour of the energy as a function of η^2 for $\beta\hbar\omega = 10$, with a linear fit for the values with $\eta < 0.5$.

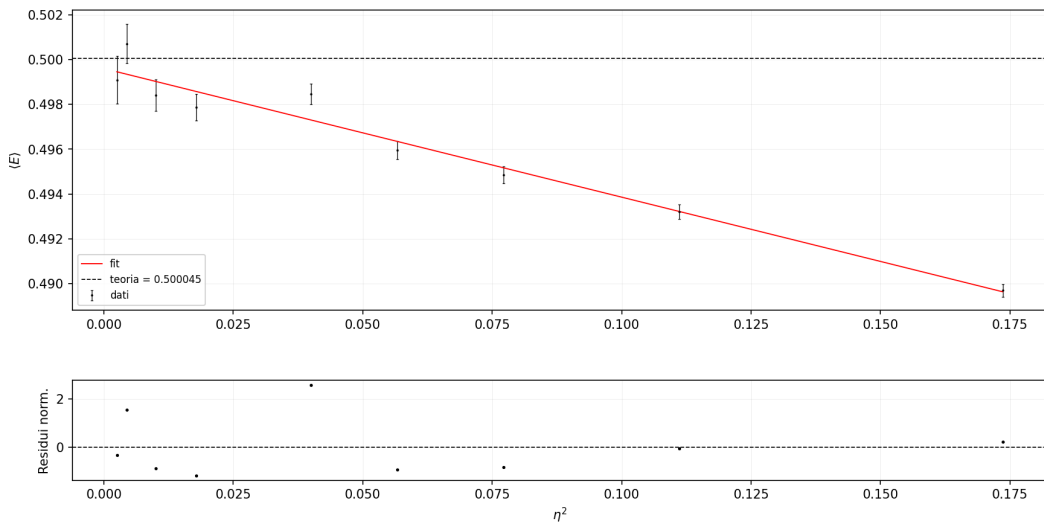


Figure 6: The same energy fit, shown with the normalised residuals, restricted to the range of η values considered in the fit.

The energy value in the continuum limit is compatible with the theoretical value of ≈ 0.5 , as can be seen from the summary table (table 2).

Table 2: Results of the two-parameter linear fit of the energy as a function of η^2 . The value of the reduced χ^2 is somewhat high, which could be due either to chance with insufficient statistics, or to the presence of non-negligible deviations of order higher than η^2 even for the largest values of η included in the fit. The value of a is compatible with the theoretical value of 0.5.

a	0.4996(3)
b	-0.0574(25)
χ^2_ν	1.864
ndf	7
$corr(a, b)$	-0.85

4.4 Energy gaps

To extract the energy gaps we resort to the effective gap which, in the dimensionless case, is defined by

$$\Delta E(n) = \frac{1}{\eta} \log \left[\frac{C(n)}{C(n+1)} \right] \quad (26)$$

Indeed, for large values of n the connected correlator behaves as

$$C(n) \sim e^{-n\eta\Delta E} \quad (27)$$

The extracted ΔE refers to the first excited level $E_{\overline{m}}$ such that

$$\langle 0 | O | \overline{m} \rangle \neq 0 \quad (28)$$

Using the definition above (equation 26) we obtain a sequence of estimates of the energy gap as a function of n , which should flatten around a plateau corresponding to the gap.

- **Gap $E_1 - E_0$:** to estimate the first energy gap we can use either the connected correlator of y or that of y^3 , but the former is more accurate, since it also satisfies

$$\langle 0 | y | n \rangle = \frac{1}{\sqrt{2}} \delta_{n,1} \quad (29)$$

whereas y^3 has non-vanishing overlap with 0 and 3 as well, so that the gap profile converges slowly to the plateau, and before it is reached in a stable way the data become very noisy (since the correlator reaches very small values and the ratio becomes unstable, with enormous uncertainties).

- **Gap $E_2 - E_0$:** to estimate the second energy gap we use the connected correlator of y^2 , which has non-vanishing overlap only with 0 and 2 (and is therefore particularly well suited as an interpolating operator for this gap).
- **Gap $E_3 - E_0$:** to estimate the third energy gap we use the connected correlator of A , but a priori this is only feasible for the harmonic oscillator, since we know its analytic solution. Indeed A (equation 18) is built ad hoc to be proportional to the Hermite polynomial H_3 , which we know to be the eigenfunction of the third excited state. It is not possible, on the other hand, to use y^3 as an interpolating operator for the third gap, since it also has non-vanishing overlap with 1.

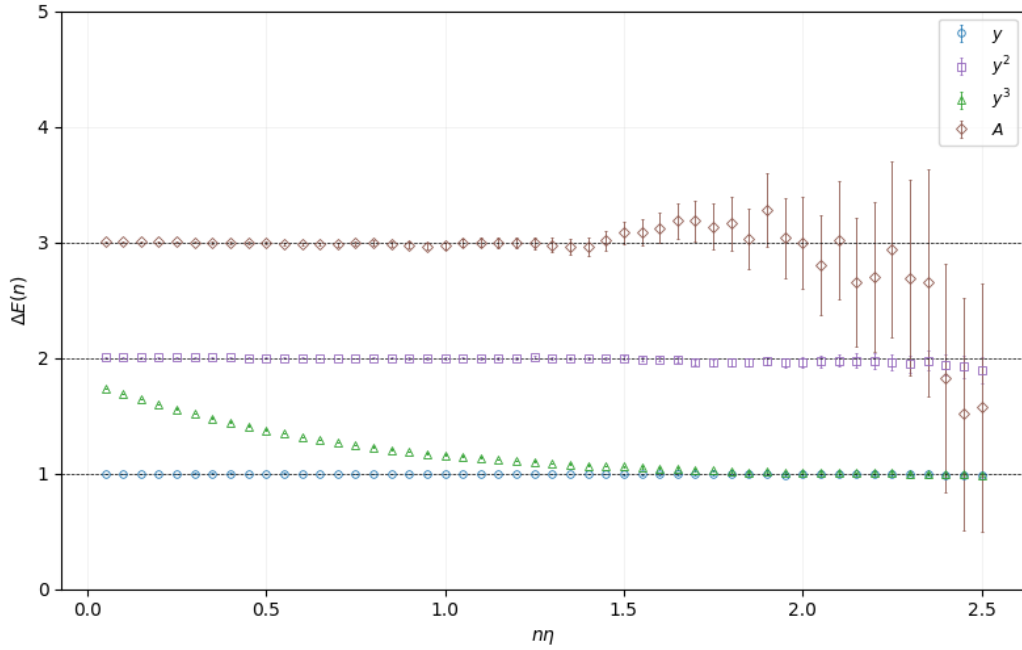


Figure 7: Effective gaps, with a clear plateau at the expected values. Data corresponding to $\beta\hbar\omega = 10$ and $N_t = 200$, removing the values with $n\eta > 2.5$, which are affected by too much noise and very large uncertainties. In the final part of the gaps relative to A a strong instability of the data begins to appear, which also characterises the tail of the other observables in the portion of data excluded from the plot.

A naive constant fit was performed on the plateau data (figure 7) in order to extract the values of the energy gaps, but it should be kept in mind that the data are correlated with one another; the fit was therefore carried out for illustrative purposes only and is not reliable for extracting precise estimates with the appropriate uncertainties. The results obtained are reported in table 3:

Table 3: Constant fit on the plateau data, performed up to $n\eta = 2.5$ (results obtained without taking into account the correlations between the data).

Energy gap	Estimate	Theoretical value
$E_1 - E_0$	0.9999(3)	1.000
$E_2 - E_0$	2.0005(8)	2.000
$E_3 - E_0$	2.999(2)	3.000

4.5 Ground-state wave function

To reconstruct the ground-state wave function we built a histogram of the sampled values of y for three combinations of $\beta\hbar\omega$ and N_t , comparing it with the theoretical probability distribution

$$P(y) = |\psi_0(y)|^2 = \frac{1}{\sqrt[4]{\pi}} e^{-y^2} \quad (30)$$

For this analysis a separate simulation was performed, using 3.5×10^4 configuration samples for each histogram and discarding the first 5×10^3 samples as a thermalisation phase.

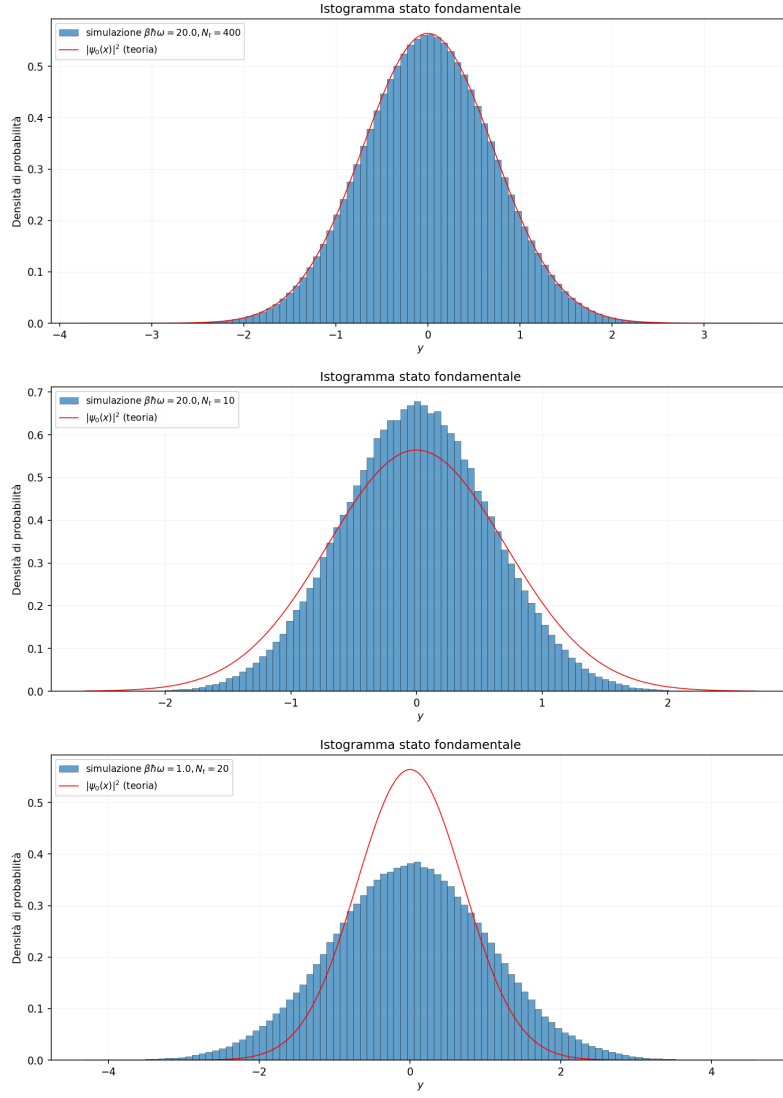


Figure 8: Histograms of the reconstruction of the ground-state wave function for three choices of parameters, compared with the theoretical probability distribution.

The only case that adequately reconstructs the ground-state distribution is the one with $\beta\hbar\omega = 20$ and $N_t = 400$ (top figure, with $\eta = 0.05$), which is close enough to the continuum limit and at a sufficiently low temperature. The case with $\beta\hbar\omega = 20$ and $N_t = 10$ (middle figure, with $\eta = 2$) does not reproduce the distribution adequately since, although the temperature is low, the time step is too large. The case with $\beta\hbar\omega = 1$ and $N_t = 20$ (bottom figure, with $\eta = 0.05$), despite the small time step, does not capture the ground state alone, since the temperature is too high to isolate it.

5 Conclusions

Given the results obtained, and knowing the analytic solution of the oscillator and its thermodynamics, we can reasonably conclude that the Markov chain implemented here samples the probability distribution of the paths correctly, since the estimates of the primary observables and of the energy gaps are compatible with the expected theoretical values, and since under the appropriate conditions we are able to adequately reconstruct the squared modulus of the ground-state wave function. By increasing the statistics one could obtain more precise results for extracting the values of the observables in the continuum limit, also allowing more accurate analyses.